Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

2020

Bachelor in Computer Applications

Course Title: **Mathematics I**

Code No: Semester: **I**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

***Note:** Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.*

Group B

2. Out of 500 people, 285 like tea, 195 like coffee, 115 like lemon juice, 45 like tea and coffee, 70 like tea and juice, 50 like juice and coffee. If 50 do not like any drinks:i) How many people like all three drinks?ii) How many people like only one drink?

3. If $x - iy = \frac{3 - 2i}{3 + 2i}$, prove that $x^2 + y^2 = 1$.

4. If H is the harmonic mean between a and b, prove that: $$\frac{1}{H - a} + \frac{1}{H - b} = \frac{1}{a + b}$$

5. Define singular and non-singular matrix. Find the inverse of the matrix A: $$A = \begin{bmatrix} 1 & -2 & -1 \\ 2 & 1 & 1 \\ 3 & -5 & 8 \end{bmatrix}$$

6. Find the focus, vertex, equation of axis, equation of directrix, and length of latus rectum of the ellipse: $$4x^2 + 9y^2 = 36$$

7. If θ is the angle between two unit vectors $\vec{a}$ and $\vec{b}$, show that: $$\left| \vec{a} - \vec{b} \right| = \sin \frac{\theta}{2}$$

8. How many numbers of at least three different digits can be formed using the digits 1, 2, 3, 4, 5, 6?

Group C

Attempt any TWO questions[2x10=20]

9. Prove by vector method: $\cos(A + B) = \cos A \cdot \cos B - \sin A \cdot \sin B$.

10. Find the equation of the circle passing through the points (1, 2), (3, 1), and (-3, -1).

11. Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 2x + 3$ and $g(x) = x^2 - 1$. Find:i) $f \circ g$ii) $f + g$iii) $g \circ f$iv) $f \circ (f \circ g)$